

Meeting Summary — Ranking Portal Development Requirements

1. Proposed Website Development Timeline & Gantt Chart

The team would like us to develop a **proposed Gantt chart for the website development timeline**. The timeline should be structured according to different development phases.

A possible phased approach discussed (NOT Explicitly Said by DORA Team, based on my intuition) is:

Phase 1 — Initial Prototype

- Develop a basic **frontend + backend prototype**.
- Implement the important/core features.
- Initially support a **single ranking** to demonstrate the complete workflow.

Phase 2 — Testing & Verification

- Test and verify the prototype with the **DORA Office team**.
- Collect feedback and identify changes or missing requirements.

Phase 3 — Multiple Rankings

- Extend the system from a single ranking to **multiple rankings**.
- Implement the required ranking-specific workflows and data structures.

Phase 4 — Detailed Features

- Develop additional/detailed features required by the institute.
- Improve the overall functionality based on feedback from earlier phases.

Phase 5 — Security & Deployment

- Focus on security-related requirements.
- Complete deployment of the system on the **actual production domains**.

Requirement: It would be better to include this proposed development timeline/Gantt chart along with the **proposal document**, if possible.

2. Technology Stack

There is **no specific technology-stack preference** from their side.

The technology stack can therefore be selected based on:

- The team's convenience and expertise.
 - The requirements of the website.
 - Scalability, maintainability, security, and deployment requirements where relevant.
-

3. Rankings to be Supported

The ranking portal is expected to manage **four different ranking systems**:

1. **NIRF**

2. NIRF contains four categories that need to be handled within the portal.

3. This is a high-priority requirement.

4. The team should try to have the NIRF functionality ready to demonstrate in the **first demo**.

5. **MDRA**

6. Best University Ranking

7. Engineering Ranking

8. **QS**

9. **THE**

The system should ultimately be designed so that the different rankings can be managed within the same ranking portal rather than requiring separate systems.

4. Public-Facing Ranking Data & Visualizations

The frontend should provide publicly viewable ranking-related information.

Year-wise Ranking Trends

The website should automatically fetch **year-wise trends/data for different rankings** from their respective ranking websites.

The purpose is to allow visitors to see how the institute's ranking/performance has changed over different years.

Comparison with Other IITs

The portal should also fetch the corresponding data of **other IITs, particularly the 2nd-generation IITs**, and provide visual comparisons.

The frontend should therefore support:

- Year-wise ranking trends.
- Data fetched from the respective ranking websites.
- Comparison with other relevant IITs.
- Visual representation of the comparison.

Priority

This is another feature that should be completed **early**, with an attempt to include it in the **first demo**.

Public Access

A normal website visitor/user should be able to visit the portal and **view this ranking data and visualizations**.

They should not need administrative or developer access to view the publicly available information.

5. Data Collection from Institute Sections

A major part of the portal will involve collecting data from different **sections, subsections, and functional/administrative units within the institute** for submission to different ranking systems.

This functionality will primarily be available to **Admin/Developer users**.

5.1 Organogram / Hierarchy

The first step should be to create an **organogram of the institute** showing the hierarchy of the different sections and subsections from which data needs to be collected.

For example, the hierarchy could include different:

- Academic sections
- Administrative sections
- Functional sections
- Subsections under these major sections

The organogram should represent the structure according to which data collection needs to be managed.

Priority: It would be better to develop this shortly after the **proposal document**.

5.2 Admin-Based Data Collection

The admin should have an option to select sections based on the institute's organogram.

For example, the admin should be able to:

1. Select a particular section/subsection.
2. Identify the data required from that section.
3. Send an email to the respective section.
4. Include a form/document through which the section can submit the required information.
5. Specify the relevant timeline/deadline for submission.

The admin should have control over **which sections receive requests for data** and when those requests are sent.

5.3 Data Submission Format

The method of collecting data does not necessarily need to be the same for every requirement.

Depending on the type of data and the requirements of the ranking document, the submission could potentially be through:

- Google Forms
- PDF
- Excel
- Other appropriate formats

For example, the different sections of the **NIRF documentation** may require different types of information & different sections (like academic, cnp, dean students) within institute, so the data collection mechanism should be selected accordingly.

5.4 Reminders & Deadlines

The system should allow administrators to manage the data-collection timeline.

If a section does not submit the requested data within the expected timeline, the admin should be able to **send reminders** to that section.

The system should therefore support tracking of:

- Which section has been asked for data.
- What data has been requested.
- The expected deadline.

- Whether the data has been received.
 - Whether a reminder needs to be sent.
-

5.5 Historical Data & Querying

Historical data should be stored in the institute's backend/database.

In particular, approximately the **previous three years of data** should be easily queryable from the backend.

This will allow the institute to retrieve and refer to historical information when preparing data for future ranking submissions.

6. Future / Later-Phase Features

Some features do not need to be implemented immediately and can be addressed in later development phases.

Historical Value Comparison During Data Entry

When a person from a particular institute section enters or updates data, the system should ideally show the **previously submitted value** alongside the new value.

For example:

Previous value: X
New value: Y
Change: Increase/Decrease

This would allow the person entering the data to immediately understand how the current value has changed compared with the previous submission.

The system could potentially indicate whether the value has:

- Increased
- Decreased
- Remained the same

This feature is intended to make data entry and verification easier, but it can be developed in a **later phase** rather than being part of the initial prototype.

7. Suggested Initial Priorities

Based on the discussion, the following features appear to have higher priority for the initial stages:

High Priority / First Demo

1. **NIRF ranking support**, including its required categories.
2. **Public frontend showing year-wise ranking trends.**
3. **Visual comparison with other IITs / relevant 2nd-generation IITs.**
4. Basic **frontend + backend prototype** demonstrating the overall workflow.
5. Institute **organogram/hierarchy** for data collection.

Immediately After Proposal / Prototype

1. Basic admin functionality for selecting sections and requesting data.
2. Data collection workflow through forms/documents.
3. Historical data storage and querying.

Later Development

1. Support and refinement for all four ranking systems.
2. Detailed data-collection workflows for different ranking requirements.
3. Automated reminders and more advanced deadline management.
4. Historical-value comparison during data entry.
5. Advanced security features.
6. Production deployment on the actual domains.

8. Key Overall Requirement

The overall objective is to develop a **centralized ranking portal** that combines:

Public Ranking Information

- Year-wise trends
- Ranking comparisons
- Comparison with other IITs
- Institute organogram

- Visual analytics

and

Internal Ranking Data Management

- Section-wise data collection
- Email-based requests
- Forms/files for data submission

- Deadlines and reminders
- Historical data storage
- Data verification and comparison

The system should initially be developed as a **working prototype with the most important features**, validated with the DORA Office team, and then progressively expanded to support all rankings and more advanced administrative, analytical, security, and deployment requirements.